Lista 1 - Processo adaptativo de sinais

Reproduza no Matlab ou Python os exemplos 3.2 e 3.6 do Livro de Adaptativos do Diniz. Produzir um relatório sobre as suas observações.

Example 3.6 An adaptive filtering algorithm is used to identify a system with impulse response given below:

$$\mathbf{h} = [0.1\ 0.3\ 0.0\ -0.2\ -0.4\ -0.7\ -0.4\ -0.2]^T$$

Consider three cases for the input signal: colored noises with variance $\sigma_x^2 = 1$ and eigenvalue spread of their correlation matrix equal to 1.0, 20, and 80, respectively. The measurement noise is Gaussian white noise uncorrelated with the input and with variance $\sigma_n^2 = 10^{-4}$. The adaptive filter has eight coefficients.

(a) Run the algorithm and comment on the convergence behavior in each case.
(b) Measure the misadjustment in each example and compare with the theoretical results where appropriate.
(c) Considering that fixed-point arithmetic is used, run the algorithm for a set of experiments and calculate the expected values for $||\Delta\mathbf{w}(k)_Q||^2$ and $\xi(k)_Q$ for the following case:

Additional noise: white noise with variance	$\sigma_n^2 = 0.0015$
Coefficient wordlength:	$b_c = 16$ bits
Signal wordlength:	$b_d = 16$ bits
Input signal: Gaussian white noise with variance	$\sigma_x^2 = 1.0$

(d) Repeat the previous experiment for the following cases:
$b_c = 12$ bits, $b_d = 12$ bits.
$b_c = 10$ bits, $b_d = 10$ bits.

(e) Suppose the unknown system is a time-varying system whose coefficients are first-order Markov processes with $\lambda_{\mathbf{w}} = 0.99$ and $\sigma_{\mathbf{w}}^2 = 0.0015$. The initial time-varying-system multiplier coefficients are the ones above described. The input signal is Gaussian white noise with variance $\sigma_x^2 = 1.0$, and the measurement noise is also Gaussian white noise independent of the input signal and of the elements of $\mathbf{n}_{\mathbf{w}}(k)$, with variance $\sigma_n^2 = 0.01$. Simulate the experiment described, measure the total excess MSE, and compare to the calculated results.

Example 3.2 Suppose in an adaptive filtering environment, the input signal consists of

$$x(k) = \cos(\omega_0 k)$$

The desired signal is given by

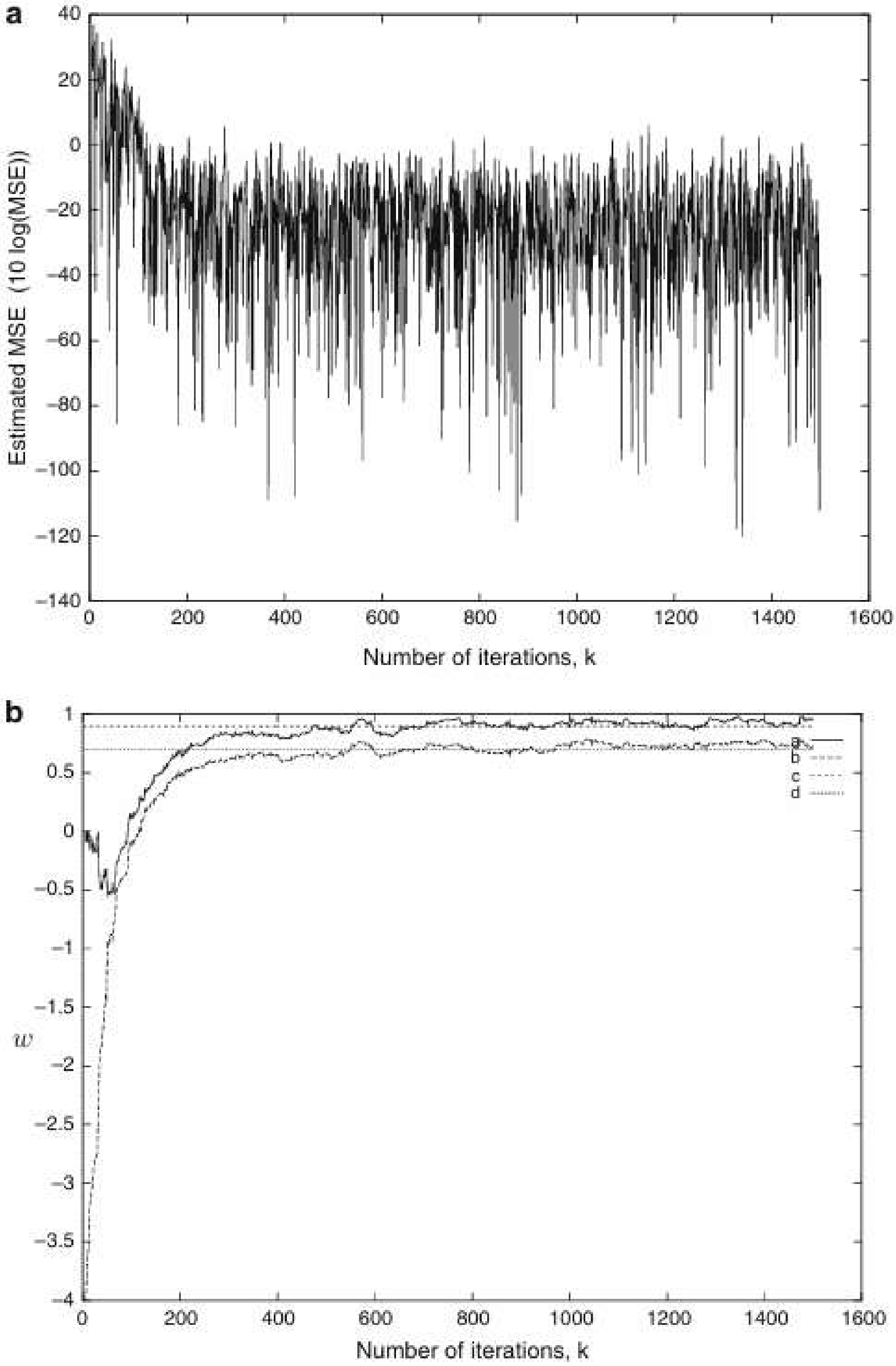

Fig. 3.5 **a** Learning curve of the instantaneous squared error, **b** Learning curves of the coefficients, a—first coefficient, b—second coefficient, c—optimal value for the first coefficient, d—optimal value of the second coefficient

$$d(k) = \sin(\omega_0 k)$$

where $\omega_0 = \frac{2\pi}{M}$. In this case $M = 7$.

Compute the optimal solution for a first-order adaptive filter.